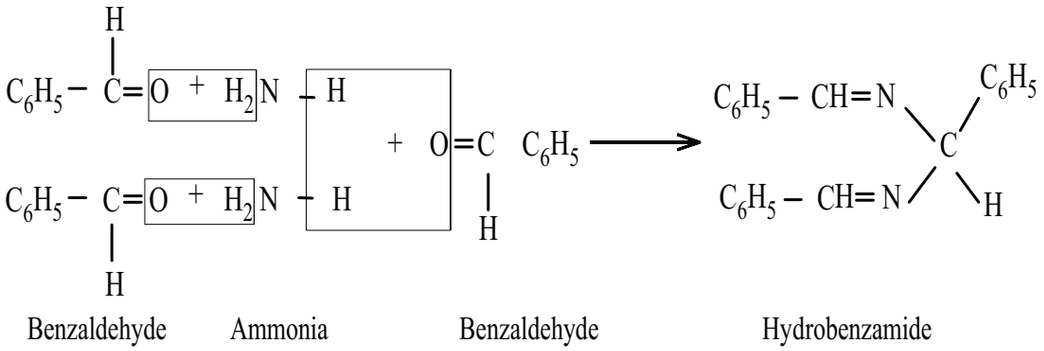
iv) Benzaldehyde form a complex condensation product with ammonia.

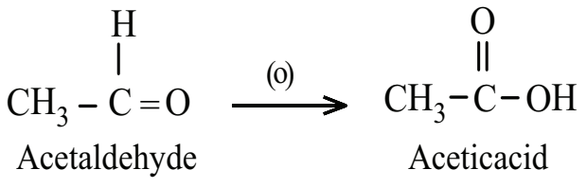


B) Oxidation of aldehydes and ketones

a) Oxidation of aldehydes

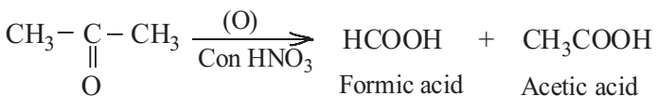
Aldehydes are easily oxidised to carboxylic acid containing the same number of carbon atom, as in parent aldehyde. The common oxidising agents are acidified $K_2Cr_2O_7$, acidic or alkaline $KMnO_4$ or chromic oxide.

Example

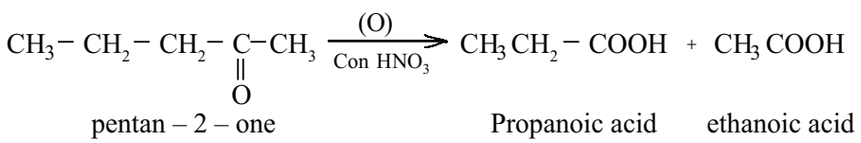


b) Oxidation of ketone

Ketones are not easily oxidised. Under drastic condition or with powerful oxidising agent like Con.HNO_3, $H^+/KMnO_4$, $H^+/K_2Cr_2O_7$, cleavage of carbon-carbon bond takes place to give a mixture of carboxylic acids having less number of carbon atom than the parent ketone.



The oxidation of unsymmetrical ketones is governed by **Popoff's rule**. It states that during the oxidation of an unsymmetrical ketone, a (C–CO) bond is cleaved in such a way that the keto group stays with the smaller alkyl group.



XII U12-Carbonyl Compounds.indd 159 08-12-2021 22:50:38